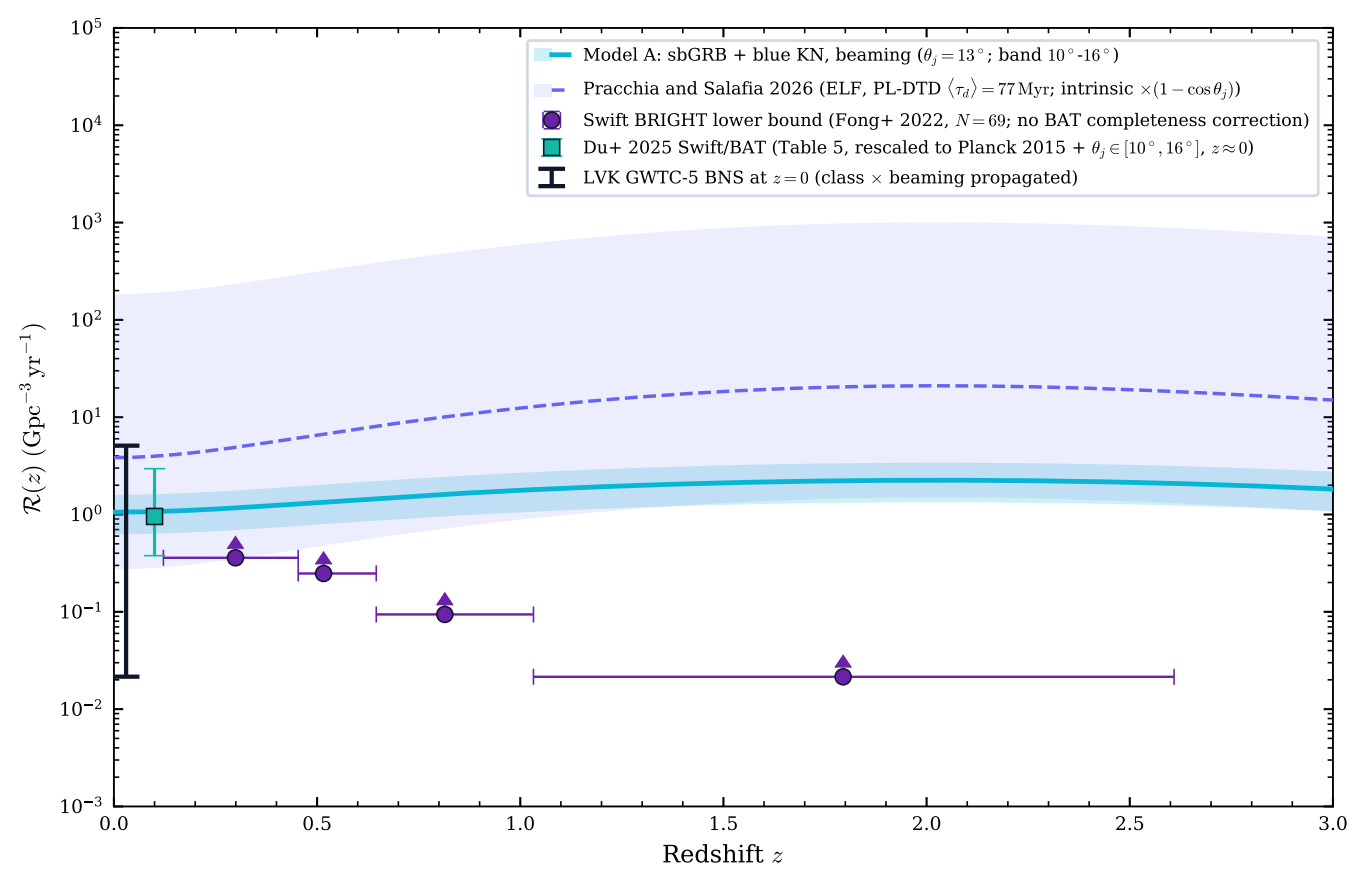

Model A: sbGRB + blue KN, beaming ($\theta_j = 13^\circ$; band 10°-16°)
Pracchia and Salafia 2026 (ELF, PL-DTD $\langle \tau_d \rangle = 77\,\mathrm{Myr}$; intrinsic $\times(1-\cos\theta_j)$)
Swift BRIGHT lower bound (Fong+ 2022, $N = 69$; no BAT completeness correction)
Du+ 2025 Swift/BAT (Table 5, rescaled to Planck 2015 + $\theta_j \in [10^\circ, 16^\circ]$, $z \approx 0$)
LVK GWTC-5 BNS at $z = 0$ (class × beaming propagated)
$\mathcal{R}(z)$ (Gpc^{-3} yr^{-1})
Redshift z
10^{5}
10^{4}
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
0.0
0.5
1.0
1.5
2.0
2.5
3.0